

IR-2 Grammar

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IR-2.1

INTRODUCTION

Chemical nomenclature may be considered to be a language. As such, it consists of words and it should obey the rules of syntax.

In the language of chemical nomenclature, the simple names of atoms are the words. As words are assembled to form a sentence, so names of atoms are assembled to form names of chemical compounds. Syntax is the set of grammatical rules for building sentences out of words. In nomenclature, syntax includes the use of symbols, such as dots, commas and hyphens, the use of numbers for appropriate reasons in given places, and the order of citation of various words, syllables and symbols.

Generally, nomenclature systems require a root on which to construct the name. This root can be an element name (*e.g.* ‘cobalt’ or ‘silicon’) for use in additive nomenclature, or can be derived from an element name (*e.g.* ‘sil’ from ‘silicon’, ‘plumb’ from ‘plumbum’ for lead) and elaborated to yield a parent hydride name (*e.g.* ‘silane’ or ‘plumbane’) for use in substitutive nomenclature.

Names are constructed by joining other units to these roots. Among the most important units are affixes. These are syllables added to words or roots and can be suffixes, prefixes or infixes according to whether they are placed after, before or within a word or root.

Suffixes and endings are of many different kinds (Table III)*, each of which conveys specific information. The following examples illustrate particular uses. They may specify the degree of unsaturation of a parent compound in substitutive nomenclature: *hexane*, *hexene*; and *phosphane*, *diphosphene*, *diphosphyne*. Other endings indicate the nature of the charge carried by the whole compound; *cobaltate* refers to an anion. Further suffixes can indicate that a name refers to a group, as in *hexyl*.

Prefixes indicate, for example, substituents in substitutive nomenclature, as in the name *chlorotrisilane*, and ligands in additive nomenclature, as in the name *aquacobalt*. Multiplicative prefixes (Table IV) can be used to indicate the number of constituents or ligands, *e.g.* *hexaaquacobalt*. Prefixes may also be used to describe the structural types or

* Tables numbered with a Roman numeral are collected together at the end of this book.

other structural features of species; geometrical and structural prefixes are listed in Table V. The ordering of prefixes in substitutive nomenclature is dealt with in Chapter IR-6, and in additive nomenclature in Chapters IR-7, IR-9 and IR-10.

Other devices may be used to complete the description of the compound. These include the charge number to indicate the ionic charge, *e.g.* hexaaquacobalt(2+), and, alternatively, the oxidation number to indicate the oxidation state of the central atom, *e.g.* hexaaquacobalt(II).

The designation of central atom and ligands, generally straightforward in mononuclear complexes, is more difficult in polynuclear compounds where there are several central atoms in the compound to be named, *e.g.* in polynuclear coordination compounds, and chain and ring compounds. In each case, a priority order or hierarchy has to be established. A hierarchy of functional groups is an established feature of substitutive nomenclature; Table VI shows an element sequence used in compositional and additive nomenclature.

The purpose of this Chapter is to guide the users of nomenclature in building the name or formula of an inorganic compound and to help them verify that the derived name or formula fully obeys the accepted principles. The various devices used in names (or formulae) are described successively below, together with their meanings and fields of application.

IR-2.2 ENCLOSING MARKS

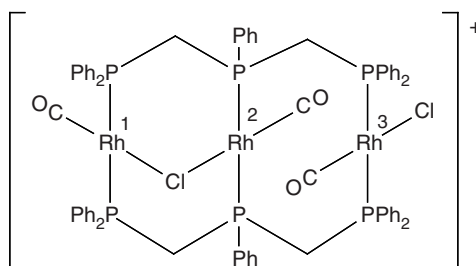
IR-2.2.1 General

Chemical nomenclature employs three types of enclosing mark, namely: braces {}, square brackets [], and parentheses ().

In *formulae*, these enclosing marks are used in the following nesting order: [], [()], [{}], [{}()], [{}()], [{}()], *etc.* Square brackets are normally used only to enclose entire formulae; parentheses and braces are then used alternately (see also Sections IR-4.2.3 and IR-9.2.3.2). There are, however, some specific uses of square brackets in formulae, *cf.* Section IR-2.2.2.1.

In *names*, the nesting order is: (), [()], {[()]}, {[()]}, *etc.* This ordering is that used in substitutive nomenclature, see Section P-16.4 of Ref. 1. (See also Section IR-9.2.2.3 for the use of enclosing marks with ligand names.)

Example:



tricarbonyl-1κC,2κC,3κC-μ-chlorido-1:2κ²Cl-chlorido-3κCl-bis{μ₃-bis[(diphenylphosphanyl)methyl]-1κP:3κP'-phenylphosphane-2κP}trirhodium(1+)

IR-2.2.2 **Square brackets**IR-2.2.2.1 *Use in formulae*

Square brackets are used in *formulae* in the following ways.

(a) To enclose the whole coordination entity of a neutral (formal) coordination compound.

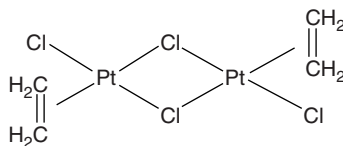
Examples:

1. $[\text{Fe}(\eta^5\text{-C}_5\text{H}_5)_2]$ (for use of the symbol η see Sections IR-9.2.4.3 and IR-10.2.5.1)
2. $[\text{Pt}(\eta^2\text{-C}_2\text{H}_4)\text{Cl}_2(\text{NH}_3)]$
3. $[\text{PH}(\text{O})(\text{OH})_2]$

No numerical subscript should follow the square bracket used in this context. For example, where the molecular formula is double the empirical formula, this should be indicated inside the square bracket.

Example:

4.



$[\{\text{Pt}(\eta^2\text{-C}_2\text{H}_4)\text{Cl}(\mu\text{-Cl})\}_2]$ is more informative than $[\text{Pt}_2(\eta^2\text{-C}_2\text{H}_4)_2\text{Cl}_4]$; the representation $[\text{Pt}(\eta^2\text{-C}_2\text{H}_4)\text{Cl}_2]_2$ is incorrect.

(b) To enclose a charged (formal) coordination entity. In this case, the superscript showing the charge appears outside the square bracket as do any subscripts indicating the number of ions in the salt.

Examples:

5. $[\text{BH}_4]^-$
6. $[\text{Al}(\text{OH})(\text{OH}_2)_5]^{2+}$
7. $[\text{Pt}(\eta^2\text{-C}_2\text{H}_4)\text{Cl}_3]^-$
8. $\text{Ca}[\text{AgF}_4]_2$
9. $[\text{Co}(\text{NH}_3)_5(\text{N}_3)]\text{SO}_4$
10. $[\text{S}_2\text{O}_5]^{2-}$
11. $[\text{PW}_{12}\text{O}_{40}]^{3-}$

(c) In a salt comprising both cationic and anionic coordination entities, each ion is separately enclosed in square brackets. (The cation is placed before the anion and no individual charges are shown.) Any subscripts indicating the number of complex ions in the salt are shown outside the square brackets.

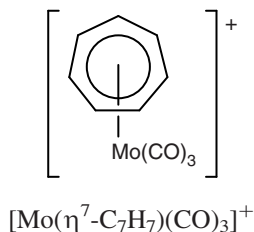
Examples:

12. $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ (comprising the ions $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{Cr}(\text{CN})_6]^{3-}$)
 13. $[\text{Co}(\text{NH}_3)_6]_2[\text{Pt}(\text{CN})_4]_3$ (comprising the ions $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{Pt}(\text{CN})_4]^{2-}$)

(d) To enclose structural formulae.

Example:

14.



(e) In solid-state chemistry, to indicate an atom or a group of atoms in an octahedral site.
 (See Section IR-11.4.3.)

Example:

15. $(\text{Mg})[\text{Cr}_2]\text{O}_4$

(f) In specifically labelled compounds (see also Section II-2.4.2.2 of Ref. 2).

Example:

16. $\text{H}_2[^{15}\text{N}]\text{NH}_2$

Note that this distinguishes the specifically labelled compound from the isotopically substituted compound $\text{H}_2^{15}\text{NNH}_2$.

(g) In selectively labelled compounds (see also Section II-2.4.3.2 of Ref. 2).

Example:

17. $[\text{}^{18}\text{O}, \text{}^{32}\text{P}]\text{H}_3\text{PO}_4$

(h) To indicate repeating units in chain compounds.

Example:

18. $\text{SiH}_3[\text{SiH}_2]_8\text{SiH}_3$

IR-2.2.2.2 Use in names

Square brackets are used in *names* in the following ways.

(a) In specifically and selectively labelled compounds the nuclide symbol is placed in square brackets before the name of the part of the compound that is isotopically modified.

(Compare with the use of parentheses for isotopically substituted compounds in Section IR-2.2.3.2, and also see Sections II-2.4.2.3, II-2.4.2.4 and II-2.4.3.3 of Ref. 2.)

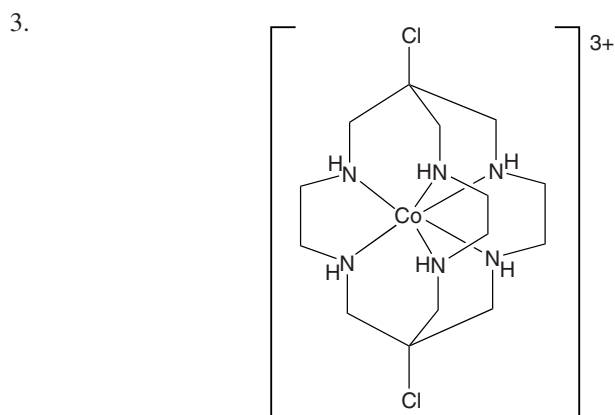
Examples:

1. $[\text{}^{15}\text{N}]\text{H}_2[\text{}^2\text{H}]$ $[\text{}^2\text{H}_1, \text{}^{15}\text{N}]\text{ammonia}$
2. $\text{HO}[\text{}^{18}\text{O}]\text{H}$ dihydrogen $[\text{}^{18}\text{O}_1]\text{peroxide}$

For more details, see Section II-2.4 of Ref. 2.

(b) When naming organic ligands and organic parts of coordination compounds the use of square brackets obeys the principles of organic nomenclature.¹

Example:

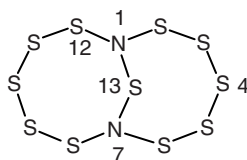


1,8-dichloro-3,6,10,13,16,19-hexaazabicyclo[6.6.6]icosanecobalt(3+)

(c) In chain and ring nomenclature, square brackets are used to enclose the nodal descriptor (Section IR-7.4.2 and Chapter II-5 of Ref. 2).

Examples:

4. $\text{HSSH}^{\bullet-}$ 1,4-dihydrony-2,3-disulfy-[4]catenate($\bullet 1-$)
- 5.



1,7-diazyundecasy-[012.1^{1,7}]dicycle

IR-2.2.3 Parentheses

IR-2.2.3.1 Use in formulae

Parentheses are used in *formulae* in the following ways.

(a) To enclose formulae for groups of atoms (the groups may be ions, substituent groups, ligands or molecules), to avoid ambiguity or when the group is being multiplied. In the latter case, a multiplicative subscript numeral follows the closing parenthesis. In the case of common ions such as nitrate and sulfate, parentheses are recommended but not mandatory.

Examples:

1. $\text{Ca}_3(\text{PO}_4)_2$
2. $[\text{Te}(\text{N}_3)_6]$
3. $(\text{NO}_3)^-$ or NO_3^-
4. $[\text{FeH}(\text{H}_2)(\text{Ph}_2\text{PCH}_2\text{CH}_2\text{PPh}_2)_2]^+$
5. $\text{PH}(\text{O})(\text{OH})_2$
6. $[\text{Co}(\text{NH}_3)_5(\text{ONO})][\text{PF}_6]_2$

(b) To enclose the abbreviation of a ligand name in formulae. (Recommended ligand abbreviations are given in Tables VII and VIII. See also Sections IR-4.4.4 and IR-9.2.3.4.)

Example:

7. $[\text{Co}(\text{en})_3]^{3+}$

(c) To enclose the superscripted radical dot and its multiplier for polyradicals, in order to avoid ambiguity in relation to multiplying the charge symbol.

Example:

8. $\text{NO}^{(2\bullet)-}$

(d) In solid-state chemistry, to enclose symbols of atoms occupying the same type of site in a random fashion. The symbols themselves are separated by a comma, with no space.

Example:

9. $\text{K}(\text{Br}, \text{Cl})$

(e) In solid-state chemistry, to indicate an atom or a group of atoms in a tetrahedral site.

Example:

10. $(\text{Mg})[\text{Cr}_2]\text{O}_4$

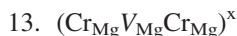
(f) To indicate the composition of a non-stoichiometric compound.

Examples:

11. $\text{Fe}_{3x}\text{Li}_{4-x}\text{Ti}_{2(1-x)}\text{O}_6$ ($x = 0.35$)
12. LaNi_5H_x ($0 < x < 66.7$)

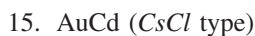
(g) In the Kröger–Vink notation (see Section IR-11.4), to indicate a complex defect.

Example:



(h) For crystalline substances, to indicate the type of crystal formed (see Chapter IR-11).

Examples:



(i) To enclose a symbol representing the state of aggregation of a chemical species.

Example:



(j) In optically active compounds, to enclose the signs of rotation.

Example:



(k) To enclose stereodescriptors, such as chirality descriptors and configuration indexes (see Section IR-9.3.3.2).

Examples:



(l) In polymers, the repeating unit is enclosed in strike-through parentheses, with the dash superimposed on the parentheses representing the bond.³

Example:



IR-2.2.3.2 Use in names

Parentheses are used in *names* in the following ways.

(a) To enclose substituent group or ligand names in order to avoid ambiguity, for example if the substituent group or ligand names contain multiplicative prefixes, such as (dioxido) or (triphosphato), or if substitution patterns would otherwise not be unambiguously specified, or if the substituent group or ligand name contains numerical or letter descriptors. It may be necessary to use different enclosing marks if the ligand names or substituent groups themselves include parentheses, *cf.* the nesting rule of Section IR-2.2.1.

Examples:

1. $[\text{Pt}(\eta^2\text{-C}_2\text{H}_4)\text{Cl}_3]^-$ trichlorido(η^2 -ethene)platinate(II)
2. $[\text{Hg}(\text{CHCl}_2)\text{Ph}]$ (dichloromethyl)(phenyl)mercury

(b) Following multiplicative prefixes of the series bis, tris, *etc.*, unless other enclosing marks are to be used because of the nesting order (see Section IR-2.2.1).

Examples:

3. $[\text{CuCl}_2(\text{NH}_2\text{Me})_2]$ dichloridobis(methylamine)copper(II)
4. Fe_2S_3 diiron tris(sulfide)

(c) To enclose oxidation and charge numbers.

Example:

5. $\text{Na}[\text{B}(\text{NO}_3)_4]$ sodium tetranitratoborate(III), or
sodium tetranitratoborate(1−)

(d) For radicals, to enclose the radical dot, and the charge number if appropriate.

Examples:

6. $\text{ClOO}^\bullet$ chloridodioxygen($\bullet$)
7. $\text{Cl}_2^{\bullet-}$ dichloride($\bullet$ 1−)

(e) To enclose stoichiometric ratios for formal addition compounds.

Example:

8. $8\text{H}_2\text{S}\cdot 46\text{H}_2\text{O}$ hydrogen sulfide—water (8/46)

(f) To enclose italic letters representing bonds between two (or more) metal atoms in coordination compounds.

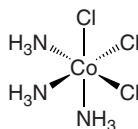
Example:

9. $[\text{Mn}_2(\text{CO})_{10}]$ bis(pentacarbonylmanganese)(*Mn—Mn*)

(g) To enclose stereochemical descriptors (see Section IR-9.3)

Examples:

10.



$[\text{CoCl}_3(\text{NH}_3)_3]$ (OC-6-22)-triamminetrichloridocobalt(III)

11. (+)₅₈₉-[Co(en)₃]Cl₃ (+)₅₈₉-tris(ethane-1,2-diamine)cobalt(III) trichloride

12.
 $(2R, 3S)\text{-Cl}\overset{1}{\text{Si}}\text{H}_2\overset{2}{\text{Si}}\text{HCl}\overset{3}{\text{Si}}\text{HCl}\overset{4}{\text{Si}}\text{H}_2\overset{5}{\text{Si}}\text{H}_3$
 (2*R*,3*S*)-1,2,3-trichloropentasilane

(h) In isotopically substituted compounds, the appropriate nuclide symbol(s) is placed in parentheses before the name of the part of the compound that is isotopically substituted (see Section II-2.3.3 of Ref. 2). Compare with the use of square brackets for specifically and selectively labelled compounds in Section IR-2.2.2.2(a).

Example:

13. H³HO (³H₁)water

(i) To enclose the number of hydrogen atoms in boron compounds.

Example:

14. B₆H₁₀ hexaborane(10)

(j) In hydrogen names (Section IR-8.4), to enclose the part of the name following the word hydrogen.

Example:

15. [HMo₆O₁₉][−] hydrogen(nonadecaoxidohexamolybdate)(1−)

IR-2.2.4 **Braces**

Braces are used in *names* and *formulae* within the hierarchical sequence outlined and exemplified in Section IR-2.2.1.

IR-2.3 **HYPHENS, PLUS AND MINUS SIGNS, ‘EM’ DASHES AND BOND INDICATORS**

IR-2.3.1 **Hyphens**

Hyphens are used in *formulae* and in *names*. Note that there is no space on either side of a hyphen.

(a) To separate symbols such as μ (mu), η (eta) and κ (kappa) from the rest of the formula or name.

Example:

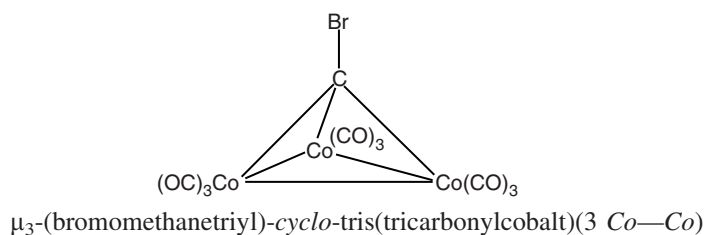
1. [{Cr(NH₃)₅}₂(μ-OH)]⁵⁺ μ-hydroxido-bis(pentaamminechromium)(5+)

(b) To separate geometrical or structural and stereochemical designators such as *cyclo*, *catena*, *triangulo*, *quadro*, *tetrahedro*, *octahedro*, *closo*, *nido*, *arachno*, *cis* and *trans* from

the rest of the formula or name. In dealing with aggregates or clusters, locant designators are similarly separated.

Example:

2.



(c) To separate locant designators from the rest of the name.

Example:

3. $\text{SiH}_2\text{ClSiHClSiH}_2\text{Cl}$ 1,2,3-trichlorotrisilane

(d) To separate the labelling nuclide symbol from its locant in the formula of a selectively labelled compound.

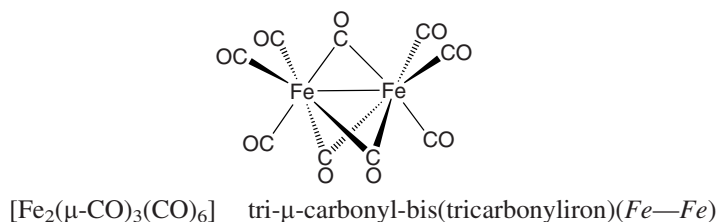
Example:

4. $[1\text{-}^2\text{H}_{1;2}]\text{SiH}_3\text{OSiH}_2\text{OSiH}_3$

(e) To separate the name of a bridging ligand from the rest of the name.

Example:

5.



IR-2.3.2

Plus and minus signs

The signs + and – are used to indicate the charge on an ion in a formula or name.

Examples:

1. Cl^-

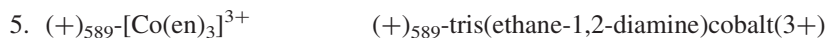
2. Fe^{3+}

3. $[\text{SO}_4]^{2-}$

4. $[\text{Co}(\text{CO})_4]^-$ tetracarbonylcobaltate(1–)

They can also indicate the sign of optical rotation in the formula or name of an optically active compound.

Example:



IR-2.3.3

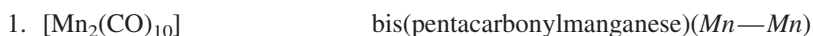
'Em' dashes

'Em' dashes are used in *formulae* only when the formulae are structural. (The less precise term 'long dashes' was used in Ref. 4.)

In *names*, 'em' dashes are used in two ways.

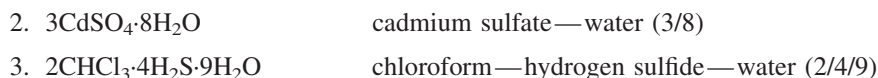
(a) To indicate metal-metal bonds in polynuclear compounds. They separate the italicized symbols of the bond partners which are contained in parentheses at the end of the name.

Example:



(b) To separate the individual constituents in names of (formal) addition compounds.

Examples:

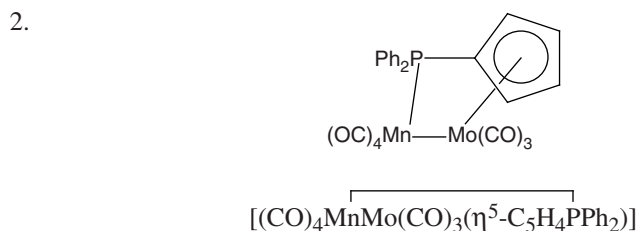
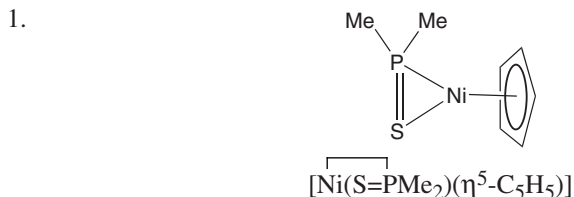


IR-2.3.4

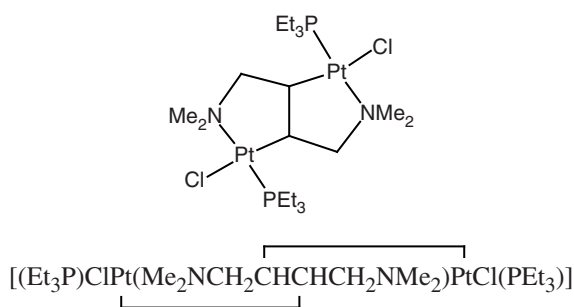
Special bond indicators for line formulae

The structural symbols $\square$ and $\square$ may be used in line formulae to indicate bonds between non-adjacent atom symbols.

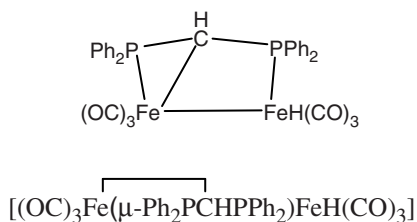
Examples:



3.



4.



IR-2.4

SOLIDUS

The solidus (/) is used in names of formal addition compounds to separate the arabic numerals which indicate the proportions of individual constituents in the compound.

Examples:

1. $\text{BF}_3 \cdot 2\text{H}_2\text{O}$ boron trifluoride—water (1/2)
2. $\text{BiCl}_3 \cdot 3\text{PCl}_5$ bismuth trichloride—phosphorus pentachloride (1/3)

IR-2.5

DOTS, COLONS, COMMAS AND SEMICOLONS

IR-2.5.1

Dots

Dots are used in *formulae* in various positions.

(a) As right superscripts they indicate unpaired electrons in radicals (see Section IR-4.6.2).

Examples:

1. $\text{HO}^\bullet$
2. $\text{O}_2^{2\bullet}$

(b) As right superscripts in the Kröger–Vink notation of solid-state chemistry, they indicate the unit of positive effective charge (see Section IR-11.4.4).

Example:

3. $\text{Li}_{\text{Li},1-2x}^x \text{Mg}_{\text{Li},x}^\bullet \text{V}_{\text{Li},x}' \text{Cl}_{\text{Cl}}^x$

(c) Centre dots in *formulae* of (formal) addition compounds, including hydrates, adducts, clathrates, double salts and double oxides, separate the individual constituents. The dot is written in the centre of the line to distinguish it from a full stop (period).

Examples:

4. $\text{BF}_3 \cdot \text{NH}_3$
5. $\text{ZrCl}_2\text{O} \cdot 8\text{H}_2\text{O}$
6. $\text{CuCl}_2 \cdot 3\text{Cu}(\text{OH})_2$
7. $\text{Ta}_2\text{O}_5 \cdot 4\text{WO}_3$

Dots are used in *names* of radicals to indicate the presence of unpaired electrons.

Examples:

8. $\text{ClO}^\bullet$ oxidochlorine(•)
9. $\text{Cl}_2^{\bullet-}$ dichloride(•1–)

IR-2.5.2

Colons

Colons are used in *names* in the following ways.

(a) In coordination and organometallic compounds, to separate the ligating atoms of a ligand which bridges central atoms.

Example:

1. $[\{\text{Co}(\text{NH}_3)_3\}_2(\mu\text{-NO}_2)(\mu\text{-OH})_2]^{3+}$
di- μ -hydroxido- μ -nitrito- $\kappa\text{N}:\kappa\text{O}$ -bis(triamminecobalt)(3+)

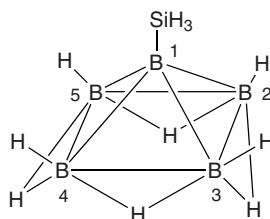
(See Sections IR-9.2.4.2 and IR-10.2.3.3 for the use of κ , and Sections IR-9.2.5.2 and IR-10.2.3.1 for the use of μ .)

(b) In polynuclear coordination and organometallic compounds, to separate the central atom locants when single ligating atoms or unsaturated groups bind to two or more central atoms. Thus, a chloride ligand bridging between central atoms 1 and 2 would be indicated by $\mu\text{-chlorido-1:2}\kappa^2\text{Cl}$, and a carbonyl group terminally bonded to atom 1 and bridging atoms 2 and 3 *via* its π electrons would be indicated by $\mu_3\text{-2}\eta^2\text{:3}\eta^2\text{-carbonyl-1}\kappa\text{C}$.

(c) In boron compounds, to separate the sets of locants of boron atoms which are connected by bridging hydrogen atoms.

Example:

2.



1-silyl-2,3:2,5:3,4:4,5-tetra- μH -pentaborane(9)

(d) In chains and rings nomenclature, to separate nodal descriptors of individual modules of an assembly (see Section IR-7.4.2).

IR-2.5.3 Commas

Commas are used in the following ways.

(a) To separate locants.

Example:



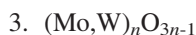
(b) To separate the symbols of the ligating atoms of a polydentate ligand.

Example:



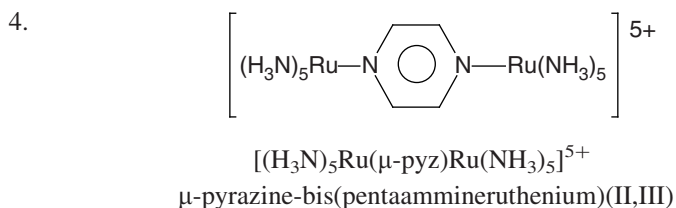
(c) In solid-state chemistry, to separate symbols of atoms occupying the same type of site in a random fashion.

Example:



(d) To separate oxidation numbers in a mixed valence compound.

Example:



(e) To separate symbols of labelled atoms in selectively labelled compounds. (See Section II-2.4.3.3 of Ref. 2.)

Example:



IR-2.5.4 Semicolons

Semicolons are used in the following ways.

(a) In the names of coordination compounds, to order locants already separated by commas, as in the kappa convention. (See examples in Section IR-9.2.5.6.)

(b) To separate the subscripts that indicate the possible numbers of labelling nuclides in selectively labelled compounds.

Example:



IR-2.6

SPACES

In inorganic nomenclature, spaces are used in *names* in the following ways in English; the rules may differ in other languages. Spaces are never used within formulae.

(a) To separate the names of ions in salts.

Examples:

- | | |
|--|------------------------------|
| 1. NaCl | sodium chloride |
| 2. NaTl(NO ₃) ₂ | sodium thallium(I) dinitrate |

(b) In names of binary compounds, to separate the electropositive part from the electronegative part.

Example:

- | | |
|-----------------------------------|---------------------------|
| 3. P ₄ O ₁₀ | tetraphosphorus decaoxide |
|-----------------------------------|---------------------------|

(c) To separate the arabic numeral from the symbols of central atoms in the bonding descriptor in the name of a polynuclear entity with several direct bonds between central atoms.

Example:

- | | |
|--|--|
| 4. [Os ₃ (CO) ₁₂] | <i>cyclo</i> -tris(tetracarbonylosmium)(3 Os—Os) |
|--|--|

(d) In names of (formal) addition compounds, to separate the stoichiometric descriptor from the remainder of the name.

Example:

- | | |
|--|-----------------------------|
| 5. 3CdSO ₄ ·8H ₂ O | cadmium sulfate—water (3/8) |
|--|-----------------------------|

(e) In solid-state nomenclature, to separate formula and structural type.

Example:

- | | |
|----------------------------------|-------------------------|
| 6. TiO ₂ (<i>o</i>) | (<i>brookite</i> type) |
|----------------------------------|-------------------------|

IR-2.7 ELISIONS

In general, in compositional and additive nomenclature no elisions are made when using multiplicative prefixes.

Example:

1. tetraaqua (*not* tetraqa)
2. monooxygen (*not* monooxygen)
3. tetraarsenic hexaoxide

However, monoxide, rather than monooxide, is an allowed exception through general use.

IR-2.8 NUMERALS

IR-2.8.1 **Arabic numerals**

Arabic numerals are crucially important in nomenclature; their placement in a formula or name is especially significant.

They are used in *formulae* in many ways.

(a) As right subscripts, to indicate the number of individual constituents (atoms or groups of atoms). Unity is not indicated.

Examples:

1. CaCl_2
2. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

(b) As a right superscript, to indicate the charge. Unity is not indicated.

Examples:

3. Cl^-
4. NO^+
5. Cu^{2+}
6. $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$

(c) To indicate the composition of (formal) addition compounds or non-stoichiometric compounds. The numeral is written on the line before the formula of each constituent except that unity is omitted.

Examples:

7. $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
8. $8\text{WO}_3 \cdot 9\text{Nb}_2\text{O}_5$

(d) To designate the mass number and/or the atomic number of nuclides represented by their symbols. The mass number is written as a left superscript, and the atomic number as a left subscript.

Examples:

9. $^{18}_8\text{O}$
10. ^3_1H

(e) As a right superscript to the symbol η , to indicate the hapticity of a ligand (see Sections IR-9.2.4.3 and IR-10.2.5.1). As a right subscript to the symbol μ , to indicate the bridging multiplicity of a ligand (see Section IR-9.2.5.2).

Example:

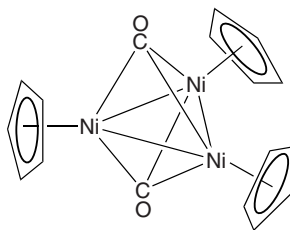
11. $[\{\text{Ni}(\eta^5\text{-C}_5\text{H}_5)\}_3(\mu_3\text{-CO})_2]$

Arabic numerals are also used as locants in *names* (see Section IR-2.14.2), and in the following ways.

(a) To indicate the number of metal–metal bonds in polynuclear compounds.

Example:

- 12.



di- μ_3 -carbonyl-*cyclo*-tris(cyclopentadienylnickel)(3 Ni—Ni)

(b) To indicate charge.

Examples:

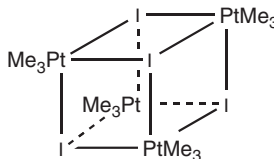
13. $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ pentaamminechloridocobalt(2+)
14. $[\text{AlCl}_4]^-$ tetrachloridoaluminate(1−)

Note that the number '1' must be included in order to avoid ambiguity in relation to symbols for optical rotation [see Section IR-2.2.3.1(j)].

(c) As a right subscript to the symbol μ , to indicate bridging multiplicity of a ligand (see Section IR-9.2.5.2).

Example:

15.



(d) In the nomenclature of boron compounds (see Chapter IR-6.2.3), to indicate the number of hydrogen atoms in the parent borane molecule. The arabic numeral is enclosed in parentheses immediately following the name.

Examples:

16. B₂H₆ diborane(6)

17. B₁₀H₁₄ decaborane(14)

(e) As a right superscript to the symbol κ , to indicate the number of donor atoms of a particular type bound to a central atom or central atoms (see Sections IR-9.2.4.2 and IR-10.2.3.3).

(f) As a right superscript to the symbol η , to indicate the hapticity of a ligand. (See Sections IR-9.2.4.3 and IR-10.2.5.1.)

(g) In polynuclear structures, arabic numerals are part of the CEP descriptor⁵ used to identify polyhedral shapes. (See also Section IR-9.2.5.6.)

(h) In the stoichiometric descriptor terminating the name of a (formal) addition compound (see Section IR-5.5).

Example:

18. 8H₂S·46H₂O hydrogen sulfide—water (8/46)

(i) As a right superscript, to indicate the non-standard bonding number in the λ convention. (See Section IR-6.2.1.)

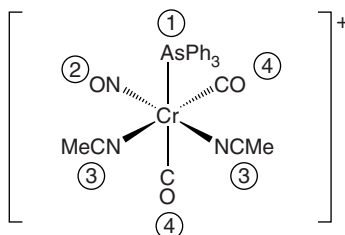
Example:

19. IH₅ λ^5 -iodane

(j) To describe the coordination geometry and configuration of ligands around a central atom using polyhedral symbols and configuration indexes (see Sections IR-9.3.2 and IR-9.3.3).

Example:

20.



(OC-6-43)-bis(acetonitrile)dicarbonylnitrosyl(triphenylarsane)chromium(1+)

IR-2.8.2

Roman numerals

Roman numerals are used in *formulae* as right superscripts to designate the formal oxidation state.

Examples:

1. $[\text{Co}^{\text{II}}\text{Co}^{\text{III}}\text{W}_{12}\text{O}_{42}]^{7-}$
2. $[\text{Mn}^{\text{VII}}\text{O}_4]^-$
3. $\text{Fe}^{\text{II}}\text{Fe}^{\text{III}}_2\text{O}_4$

In *names* they indicate the formal oxidation state of an atom, and are enclosed in parentheses immediately following the name of the atom being qualified.

Examples:

4. $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ hexaaquairon(II)
5. $[\text{FeO}_4]^{2-}$ tetraoxido ferrate(VI)

IR-2.9

ITALIC LETTERS

Italic letters are used in *names* as follows.

- (a) For geometrical and structural prefixes such as *cis*, *cyclo*, *catena*, *triangulo*, *nido*, *etc.* (see Table V).
- (b) To designate symbols of central atoms in the bonding descriptor in polynuclear compounds.

Example:

1. $[\text{Mn}_2(\text{CO})_{10}]$ bis(pentacarbonylmanganese)(*Mn—Mn*)

- (c) In double oxides and hydroxides when the structural type is to be indicated.

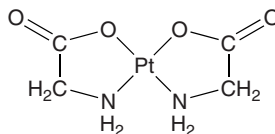
Example:

2. MgTiO_3 (*ilmenite* type)

(d) In coordination compounds, to designate the symbols of the atom or atoms of a ligand (usually polydentate) to which the central atom is bound, whether the kappa convention is used or not. (See Section IR-9.2.4.4.)

Example:

3.



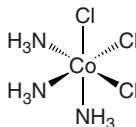
cis-bis(glycinato- $\kappa N, \kappa O$)platinum

(e) In solid-state chemistry, in Pearson and crystal system symbols. (See Sections IR-3.4.4 and IR-11.5.)

(f) Italicized capital letters are used in polyhedral symbols. (See Section IR-9.3.2.1.)

Example:

4.



$[\text{CoCl}_3(\text{NH}_3)_3]$ (OC-6-22)-triamminetrichloridocobalt(III)

(g) Other uses of italicized capital letters are as locants in substitutive nomenclature (see, for example, Section IR-6.2.4.1), and the letter *H* for indicated hydrogen (see, for example, Section IR-6.2.3.4). Italic lower case letters are used to represent numbers, especially in formulae where the numbers are undefined.

Examples:

5. $(\text{HBO}_2)_n$

6. Fe^{n+}

IR-2.10

GREEK ALPHABET

Greek letters (in Roman type) are used in systematic inorganic nomenclature as follows:

- Δ to show absolute configuration, or as a structural descriptor to designate deltahedra (see Section IR-9.3.4);
- δ to denote the absolute configuration of chelate ring conformations (see Section IR-9.3.4); in solid-state chemistry to indicate small variations of composition (see Section IR-11.3.2); to designate cumulative double bonds in rings or ring systems (see Section P-25.7 of Ref. 1);
- η to designate the hapticity of a ligand (see Sections IR-9.2.4.3 and IR-10.2.5.1);

- κ as a ligating atom designator in the kappa convention (see Sections IR-9.2.4.2 and IR-10.2.3.3);
- Λ to show absolute configuration (see Section IR-9.3.4);
- λ to indicate non-standard bonding number in the lambda convention (see Section IR-6.2.1 and Section P-14.1 of Ref.1); to denote the absolute configuration of chelate ring conformations (see Section IR-9.3.4);
- μ to designate a bridging ligand (see Sections IR-9.2.5.2 and IR-10.2.3.1).

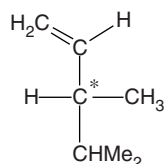
IR-2.11 ASTERISKS

The asterisk (*) is used in *formulae* as a right superscript to the symbol of an element, in the following ways:

- (a) To highlight a chiral centre.

Example:

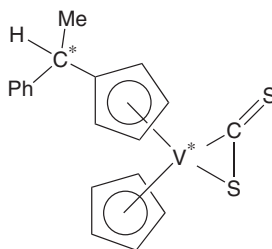
1.



This usage has been extended to label a chiral ligand or a chiral centre in coordination chemistry.

Example:

2.



- (b) To designate excited molecular or nuclear states.

Example:

3. NO*

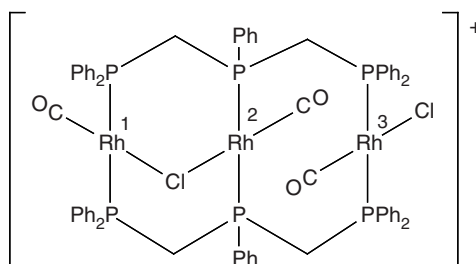
IR-2.12 PRIMES

(a) Primes ('), double primes (''), triple primes ('''), *etc.* may be used in the names and formulae of coordination compounds in the following ways:

- (i) within ligand names, in order to differentiate between sites of substitution;
- (ii) when specifying donor atoms (IR-9.2.4.2), in order to differentiate between donor atoms;

- (iii) when specifying configuration using configuration indexes (IR-9.3.5.3), in order to differentiate between donor atoms of the same priority, depending on whether they are located within the same ligand or portion of the ligand.

Example:



tricarbonyl-1 κ C,2 κ C,3 κ C- μ -chlorido-1:2 κ^2 Cl-chlorido-3 κ Cl-bis{ μ_3 -bis[(diphenylphosphanyl)methyl]-1 κ P:3 κ P'-phenylphosphane-2 κ P}trirhodium(1+)

- (b) Primes, double primes, triple primes, *etc.* are also used as right superscripts in the Kröger–Vink notation (see Section IR-11.4) where they indicate a site which has one, two, three, *etc.* units of negative effective charge.

Example:



IR-2.13 MULTIPLICATIVE PREFIXES

The number of identical chemical entities in a name is expressed by a multiplicative prefix (see Table IV).

In the case of simple entities such as monoatomic ligands the multiplicative prefixes di, tri, tetra, penta, *etc.*, are used.

The multiplicative prefixes bis, tris, tetrakis, pentakis, *etc.* are used with composite ligand names or in order to avoid ambiguity. The modified entity is placed within parentheses.

Examples:

- | | |
|--|--|
| 1. Fe_2O_3 | diiron trioxide |
| 2. $[\text{PtCl}_4]^{2-}$ | tetrachloridoplatinate(2-) |
| 3. $[\text{Fe}(\text{CCPh})_2(\text{CO})_4]$ | tetracarbonylbis(phenylethynyl)iron |
| 4. TlI_3 | thallium tris(iodide) (<i>cf.</i> Section IR-5.4.2.3) |
| 5. $\text{Ca}_3(\text{PO}_4)_2$ | tricalcium bis(phosphate) |
| 6. $[\text{Pt}(\text{PPh}_3)_4]$ | tetrakis(triphenylphosphane)platinum(0) |

Composite multiplicative prefixes are built up by citing units first, then tens, hundreds and so on, *e.g.* 35 is written pentatriaconta (or pentatriacontakis).

IR-2.14 LOCANTS

IR-2.14.1 Introduction

Locants are used to indicate the position of a substituent on, or a structural feature within, a parent molecule. The locants can be arabic numerals or letters.

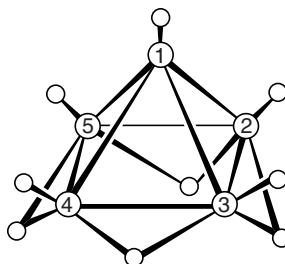
IR-2.14.2 Arabic numerals

Arabic numerals are used as locants in the following ways.

(a) For numbering skeletal atoms in parent hydrides, to indicate: the placement of hydrogen atoms when there are non-standard bonding numbers; unsaturation; the positions of bridging hydrogen atoms in a borane structure.

Examples:

1. $\text{H}_5\overset{1}{\text{S}}\overset{2}{\text{S}}\overset{3}{\text{S}}\overset{4}{\text{S}}\text{H}$ $1\lambda^6,3\lambda^6$ -tetrasulfane (not $2\lambda^6,4\lambda^6$)
2. $\text{H}_2\overset{1}{\text{N}}\overset{2}{\text{N}}=\overset{3}{\text{N}}\overset{4}{\text{N}}\overset{5}{\text{N}}\text{H}_2$ pentaaz-2-ene
- 3.



2,3:2,5:3,4:4,5-tetra- μ H-nido-pentaborane(9)

(b) In replacement nomenclature.

Example:

4. $\overset{1}{\text{CH}_3}\overset{2}{\text{S}}\overset{3}{\text{CH}_2}\overset{4}{\text{SiH}_2}\overset{5}{\text{CH}_2}\overset{6}{\text{CH}_2}\overset{7}{\text{O}}\overset{8}{\text{CH}_2}\overset{9}{\text{CH}_2}\overset{10}{\text{O}}\overset{11}{\text{CH}_3}$
7,10-dioxa-2-thia-4-silaundecane

(c) In additive nomenclature.

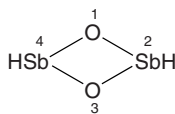
Example:

5. $\overset{1}{\text{SiH}_3}\overset{2}{\text{GeH}_2}\overset{3}{\text{SiH}_2}\overset{4}{\text{SiH}_2}\overset{5}{\text{SiH}_3}$
1,1,1,2,2,3,3,4,4,5,5,5-dodecahydrido-2-germy-1,3,4,5-tetrasilyl-[5]catena

(d) In the Hantzsch–Widman nomenclature (Section IR-6.2.4.3), to indicate positions of skeletal atoms.

Example:

6.

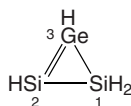


1,3,2,4-dioxadistibetane

(e) In the Hantzsch–Widman nomenclature (Section IR-6.2.4.3), to denote indicated hydrogen.

Example:

7.

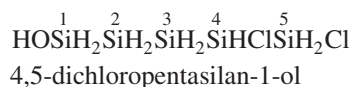


1H-1,2,3-disilagermirene

(f) In substitutive nomenclature, to specify the positions of substituent groups.

Example:

8.



(g) In substitutive nomenclature, to specify the skeletal atom at which an additive or subtractive operation is performed.

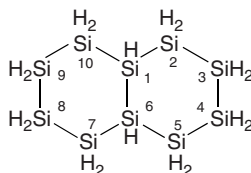
Example:

9. •HNNH• and –HNNH– hydrazine-1,2-diyl

(h) In von Baeyer names, to indicate the topology of a polycyclic ring system.

Example:

10.



bicyclo[4.4.0]decasilane

(i) In polynuclear coordination compounds, for numbering the central atoms (see Section IR-9.2.5).

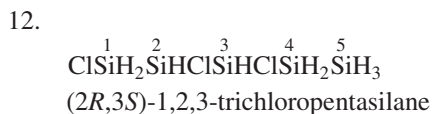
Example:

11.



(j) To indicate stereochemistry at particular atoms in structures where arabic numerals have been used for numbering those atoms.

Example:



IR-2.14.3 Letter locants

Italicized upper case letters are used as locants in certain substitutive names. (See, for example, Section IR-6.2.4.1.)

Lower case letters are used in polyoxometallate nomenclature to designate the vertices of the coordination polyhedra around the central atoms. They are attached to the number of the central atom to which a particular vertex refers. A detailed treatment is given in Chapter II-1 of Ref. 2.

IR-2.15 ORDERING PRINCIPLES

IR-2.15.1 Introduction

Chemical nomenclature deals with names of elements and their combinations. Whereas writing the symbol or the name of an element is straightforward, a choice of which element to write first in the formula and in the name has to be made as soon as an element is associated with one or more other elements to form, for example, a binary compound. The order of citation of elements in formulae and names is based upon the methods outlined below. Furthermore, groups of atoms, such as ions, ligands in coordination compounds and substituent groups in derivatives of parent hydrides, are ordered according to specified rules.

IR-2.15.2 Alphabetical order

Alphabetical order is used in *formulae* as follows.

(a) Within the group of cations and within the group of anions, respectively, in formulae of salts and double salts. Deviations from this rule may be acceptable if it is desired to convey specific structural information, as in Example 5 below.

Examples:

1. BiClO (anions Cl⁻ and O²⁻)
2. NaOCl (the anion is OCl⁻, *cf.* Section IR-4.4.3.1)
3. KNa₄Cl(SO₄)₂
4. CaTiO₃ (*perovskite* type)
5. SrFeO₃ (*perovskite* type)

(b) In formulae of coordination compounds and species formally treated as coordination compounds, ligands are cited in alphabetical order of their formulae or abbreviations

(*cf.* Section IR-2.15.3.4). Where possible, the donor atom symbol in ligand formulae should be placed nearest the symbol of the central atom to which it is coordinated. (See Section IR-9.2.3.1.)

Example:



(c) In the construction of the formula for a (formal) addition compound, the formulae of the individual components are ordered first by number of each component, then alphabetically. (See Section IR-4.4.3.5.)

Alphabetical order is used in *names* as follows.

(d) In compositional names, the names of the formally electropositive components and the names of the formally electronegative components are each arranged alphabetically with the former group of components preceding the latter. Note that this order of components may therefore deviate from the order of the corresponding components in the formula, as in Examples 7, 9 and 10 below.

Examples:

7. KMgF_3	magnesium potassium fluoride
8. BiClO	bismuth chloride oxide
9. $\text{ZnI}(\text{OH})$	zinc hydroxide iodide
10. SrFeO_3	iron strontium oxide (<i>cf.</i> Example 5 above)

(e) In the citation of ligands in additive names. The alphabetical citation of ligand names is maintained regardless of the number of each ligand, or whether the compounds are mononuclear or polynuclear (*cf.* Section IR-2.15.3.4).

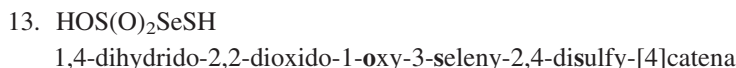
Examples:

11. $\text{K}[\text{AuS}(\text{S}_2)]$	potassium (disulfido)sulfidoaurate(1–)
12. $[\text{CrCl}_2(\text{NH}_3)_4]^+$	tetraamminedichloridochromium(1+)

A similar rule applies when citing names of substituent groups in substitutive nomenclature (see Section IR-6.3.1).

(f) For citation of the names of the skeletal atoms in the chains and rings additive nomenclature (*cf.* Section IR-7.4.3).

Example:



(g) In the construction of the name for a (formal) addition compound, the names of the individual components are ordered first by number of each component, then alphabetically. (See Section IR-5.5.)

IR-2.15.3 **Other ordering rules**IR-2.15.3.1 *Element ordering on the basis of the periodic table*

One important element sequence based on the periodic table is shown in Table VI. The element columns (1 to 18) are connected by arrows leading in a direction starting from the less metallic elements and moving towards the more metallic elements. Only H has a unique position. This order has its origin in electronegativity considerations even though O is now placed at its usual position in group 16. It is used for ordering element symbols and element names in the following cases.

(a) In compositional names of binary compounds and corresponding formulae, the element encountered *last* when following the arrow in Table VI is represented *first* in the formula as well as the name.

Examples:

1. S_2Cl_2 disulfur dichloride
2. O_2Cl dioxygen chloride
3. H_2Te dihydrogen telluride
4. AlH_3 aluminium trihydride

(b) In additive names of polynuclear compounds, the central atom encountered *last* when following the arrow is listed first, *cf.* Sections IR-7.3.2 and IR-9.2.5.1.

(c) In additive names for chains and rings, to determine the numbering of the skeletal atoms if this is not defined fully by the structure of the skeleton. The element encountered *first* when following the arrows in Table VI is given the *lowest* number. Note that the element 'y' terms (Table X) are cited alphabetically.

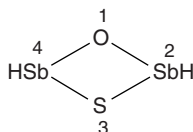
Example:

5. $\text{HOS(O)}_2\text{SeSH}$
1,4-dihydro-2,2-dioxido-1-oxy-3-seleny-2,4-disulfo-[4]catena

(d) In Hantzsch–Widman names, the element encountered *first* when following the arrows in Table VI is given the *lowest* number. The element 'a' terms (Table X) are cited in the same order.

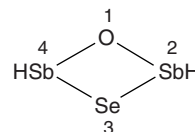
Examples:

6.



1,3,2,4-oxathiadistibetane

7.

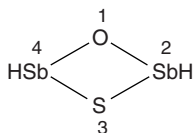


1,3,2,4-oxaselenadistibetane

(e) In names where heteroatoms replacing skeleton atoms are indicated by 'a' prefixes, the element encountered *first* when following the arrows in Table VI is given the *lowest* number. The element 'a' terms (Table X) are cited in the same order.

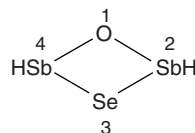
Examples:

8.



1-oxa-3-thia-2,4-distibacyclobutane

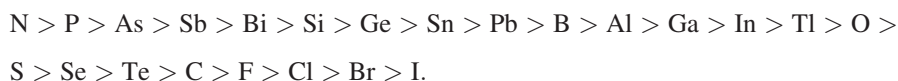
9.



1-oxa-3-selena-2,4-distibacyclobutane

IR-2.15.3.2 Ordering of parent hydrides

Where there is a choice of parent hydrides among those listed in Table IR-6.1 (or corresponding hydrides with non-standard bonding numbers, *cf.* Section IR-6.2.2.2), the name is based on the parent hydride of the element occurring first in the sequence:



This applies in particular to the naming of organometallic compounds of elements of groups 13–16 when a choice has to be made between several parent hydrides (Section IR-10.3.4).

Example:



Note that due to the rules of substitutive nomenclature¹ the above does not necessarily come into play even if two or more elements appearing in the sequence are present in the compound. For example, the substitutive name for HTeOH is tellanol, *i.e.* based on tellane, not oxidane, because the characteristic group OH must be cited as a suffix.

IR-2.15.3.3 Ordering characteristic groups for substitutive nomenclature

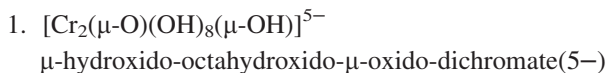
In substitutive nomenclature, an order for the choice of principal functional group is defined (see Section P-41 of Ref.1).

IR-2.15.3.4 Ordering ligands in formulae and names

In *formulae* of coordination compounds, the formulae or abbreviations representing the ligands are cited in alphabetical order as the general rule. Bridging ligands are cited immediately after terminal ligands of the same kind, if any, and in increasing order of bridging multiplicity. (See also Sections IR-9.2.3 and IR-9.2.5.)

In *names* of coordination compounds, the ligand names precede the name of the central atom and are cited in alphabetical order. For each ligand type, bridging ligands are cited immediately before terminal ligands of the same kind, if any, *e.g.* di- μ -chlorido-tetrachlorido, and in decreasing order of bridging multiplicity, *e.g.* μ_3 -oxido-di- μ -oxido... (See also Sections IR-9.2.2 and IR-9.2.5.1.)

Example:



Thus, for both *formulae* and *names* the terminal ligands are closer to the central atom, with the multiplicity of the bridging ligands increasing further away from the metal.

IR-2.15.3.5 *Ordering components in salt formulae and names*

In formulae and names of salts, double salts and coordination compounds, cations precede anions. Ordering within each of these groups is alphabetical, *cf.* Section IR-2.15.2.

IR-2.15.3.6 *Isotopic modification*

In isotopically modified compounds, a principle governs the order of citation of nuclide symbols. (See Section II-2.2.5 of Ref. 2.)

IR-2.15.3.7 *Stereochemical priorities*

In the stereochemical nomenclature of coordination compounds, the procedure for assigning priority numbers to the ligating atoms of a mononuclear coordination system is based upon the standard sequence rules developed for chiral carbon compounds (the Cahn, Ingold, Prelog or CIP rules⁶, see Section IR-9.3.3.2).

IR-2.15.3.8 *Hierarchical ordering of punctuation marks*

In the names of coordination compounds and boron compounds, the punctuation marks used to separate the symbols of atoms from the numerical locants, the locants indicating bridging atoms, and the various other sets of locants which may be present, are arranged in the following hierarchy:

semicolon > colon > comma.

The colon is only used for bridging ligands, so that the more restricted general hierarchy is simply comma < semicolon. The sequence when bridging ligands are being specified is comma < colon. (See Example 2 in Section IR-2.5.2, and Section IR-9.2.5.5.)

IR-2.16 FINAL REMARKS

In this chapter, the various uses of letters, numerals and symbols in names and formulae have been gathered under common headings to provide an easy check to ensure that the constructed name or formula is in accord with agreed practice. However, this chapter is not sufficient to make clear all the rules needed to build a name or a formula, and the reader is therefore advised to consult other appropriate chapters for the more detailed treatment.

IR-2.17 REFERENCES

1. *Nomenclature of Organic Chemistry, IUPAC Recommendations*, eds. W.H. Powell and H. Favre, Royal Society of Chemistry, in preparation: [See also, *Nomenclature of Organic Chemistry*, Pergamon Press, Oxford, 1979; *A Guide to IUPAC Nomenclature of Organic Compounds, Recommendations 1993*, eds. R. Panico, W.H. Powell and J.-C. Richer, Blackwell Scientific Publications, Oxford, 1993; and corrections in *Pure Appl. Chem.*, **71**, 1327–1330 (1999)].
2. *Nomenclature of Inorganic Chemistry II, IUPAC Recommendations 2000*, eds. J.A. McCleverty and N.G. Connelly, Royal Society of Chemistry, 2001. (Red Book II.)
3. *Compendium of Macromolecular Nomenclature*, ed. W.V. Metanowski, Blackwell Scientific Publications, Oxford, 1991. (The Purple Book. The second edition is planned for publication in 2005).
4. *Nomenclature of Inorganic Chemistry, IUPAC Recommendations 1990*, ed. G.J. Leigh, Blackwell Scientific Publications, Oxford, 1990.
5. J.B. Casey, W.J. Evans and W.H. Powell, *Inorg. Chem.*, **20**, 1333–1341 (1981).
6. R.S. Cahn, C. Ingold and V. Prelog, *Angew. Chem., Int. Ed. Engl.*, **5**, 385–415 (1966); V. Prelog and G. Helmchen, *Angew. Chem., Int. Ed. Engl.*, **21**, 567–583 (1982).